Phu Nguyen

Before this lab I thought a more complex model would always fit better. The plots showed that’s not true. The degree 1 line was too simple and missed the sine wave, and the degree 15 line got really wiggly and followed the noisy points. Degree 4 followed the wave without bending to every point, so it was the best fit. The learning curves helped me most. For degree 1, the training and cross-validation scores were both low and close together, which means high bias. For degree 15, the training score was high but the cross-validation score was much lower, which means high variance. This lab helped me see that a model needs to be complex enough to catch the pattern but not so complex that it memorizes the noise.

Cyndy Ndong

Doing this lab helped me really understand how a machine learning project works from start to finish. At first, it felt like I was just following steps without knowing why they mattered. But as I worked through it, I saw how every part connects. Cleaning the data, scaling the numbers, training the model, and checking the results all depend on each other. If you skip a step, the model just won't work right.

I also learned that accuracy isn't the only thing that matters. Understanding precision and recall showed me why false alarms and missed predictions matter in real life. High precision is important when a false alarm causes trouble, like accidentally sending a good email to the spam folder. High recall is important when you can't afford to miss something, like a doctor using AI to catch a serious disease. It depends on what problem you are trying to solve.

Another big lesson was learning about cross validation. Splitting your data just once can give you an overly lucky or unlucky score depending on how the data was shuffled. Testing the model five different times and averaging the scores gives you a much fairer and more honest idea of how well it works.

When my code broke because of missing files or wrong column names, fixing those errors helped me learn better. I had to look closely at my code and figure out what went wrong. Overall, I feel much more confident about how machine learning works now